Harish A

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Summary:

Software Engineer with 3+ years of experience in backend development, RESTful microservices, and cloud-native web applications. Currently at Google contributing to large-scale streaming and Dynamic Ad Insertion platforms; previously delivered distributed systems at Cognizant for enterprise clients including Microsoft and Verizon.

Skilled in Java, Python, Linux, CI/CD, Docker, Kubernetes, PostgreSQL/MySQL, cloud platforms (AWS, Azure, GCP), monitoring, automation, and production incident management. Focused on building scalable, secure, and high-availability systems using modern engineering best practices.

Skills:

Programming Languages:
- Java, Python, Javascript, SQL

Frameworks/Technologies:
- SpringBoot, Node.js, Express.js, Flask, FastAPI
- gRPC, Kafka

Web Technologies:
- HTML5, CSS3, Svelte

Database/Cache:
- MySQL, MongoDB, Redis

Cloud/DevOps:
- AWS (EC2, S3, Lambda, Athena)
- Microsoft Azure (Azure Functions, AKS, Event Hub)
- Docker, Kubernetes

Software Development Skills:
- Cloud Computing, Data Structures and Algorithms
- Microservice Architecture, RESTful APIs

Work Experience:

Google

Partner Solution Engineer – Video Streaming
(May 2025 – Present)

Key Contributions:
- Provide technical consultation and architectural guidance to leading OTT publishers including Disney+, DAZN, and SonyLiv on Dynamic Ad Insertion (DAI) and SSAI solutions.
- Support Google's SSAI platform integrated with Google Ad Manager for VAST/VMAP compliant ad decisioning and delivery.
- Troubleshoot and optimize Live & VOD streaming workflows using HLS and MPEG-DASH protocols for high-profile live events with millions of concurrent viewers.
- Conduct production incident triage, log analysis, performance bottleneck identification, and

root cause analysis during live events.
- Design and develop real-time monitoring dashboards to track:
  - Ad fill rates
  - Latency metrics
  - CDN performance
  - Error rates
  - Bitrate switching patterns
- Automate operational workflows and monitoring alerts to proactively detect anomalies during large-scale events.
- Work closely with product engineering, and customer teams globally to enhance reliability, scalability, and security posture.

Cognizant Technology Solutions Corp

Associate

(Oct 2024 - Present)

Client Project: Verizon

Role: Software Developer

Technologies Used: Java, SpringBoot, AWS, Spring Gateway

Key Contributions:

- Developed and Optimised RESTful API's for OSS (Operations Support System), including Trouble ticket management, Usage management using Java SpringBoot with the newer and robust microservice architecture for performance improvement and scalability.
- Built a scalable and configurable API Gateway using Spring Cloud Gateway for customised routing across multiple microservices based on various parameters improving system flexibility. • Collaborated with various on-site and client teams to thoroughly understand the requirements and deliver the solutions with attention to detail.

Client Project: DirecTV Billing & Payments

Role: Software Developer

Technologies Used: Java, Apache Spark, Hive, Athena, Shell Scripting

Key Contributions:

- Developed a new feature for billing & payments big data pre-processors and use Apache Spark for map-reduce operations on large datasets.
- Worked on a feature development for DirecTV Supplier Management System for managing contracts, channels, subscriptions.
- Automated data copying and test data creation process using shell scripting, reducing manual effort.

Programmer Analyst

(Sep 2022 - Sep 2024)

Client Project: Microsoft

Role: Software Consultant

Technologies Used: Javascript, node.js, express.js, postgreSQL, Azure Functions, Event Hub

Key Contributions:

- Build multi-cloud adapter using node.js for Media CMS(Content Management System) backend services for seamless integration between cloud providers, reducing cost and increasing flexibility and performance.
- Developed Event driven systems for content publishing, scheduled publishing, unpublishing assets with expired license and other metadata management services.
- Worked on Bulk Operation services for large data ingestion and publishing, ensuring efficient

processing of high-volume metadata.

- Collaborated on multiple debugging sessions, leveraging a deep understanding of the system to provide sustainable solutions and address issues effectively, ensuring long-term stability and performance.

Internal Project: Cognizant ISG - Automated Advertisement Classification

Role: Software Developer

Technologies Used: Python, GCP, Cloud Functions, React.js

Key Contributions:

- Designed end-to-end system architecture for automated advertisement classification on the GCP stack using Cloud Functions and Cloud Storage for and event driven design pattern, mongoDB for storing the results.
- Contributed significantly to prompt engineering for video and frame analysis using Generative AI. This involved writing multiple prompts, optimising the input prompts by extracting accurate insights from previous results, and tailoring the prompts for different scenarios based on the content.
- Developed a UI using React.js framework for uploading, analysing and reviewing the results, ensuring responsive front-end experience.
- Tested the solution using client data, achieving accuracy rates above 85% and reducing the manual efforts by over 50% by using parallel processing combined with manual validation to enhance accuracy.

Education:

Amrita School of Engineering, Bangalore

Bachelor of Technology, Electronics and Communications Engineering

2018 - 2022

Licenses & Certifications:

Certified Cloud Practitioner - Amazon Web Services (AWS)

GCP Cloud Digital Leader - Google Cloud